DHCP SERVER

(DYNAMIC HOST CONFIGURATION PROTOCOL)

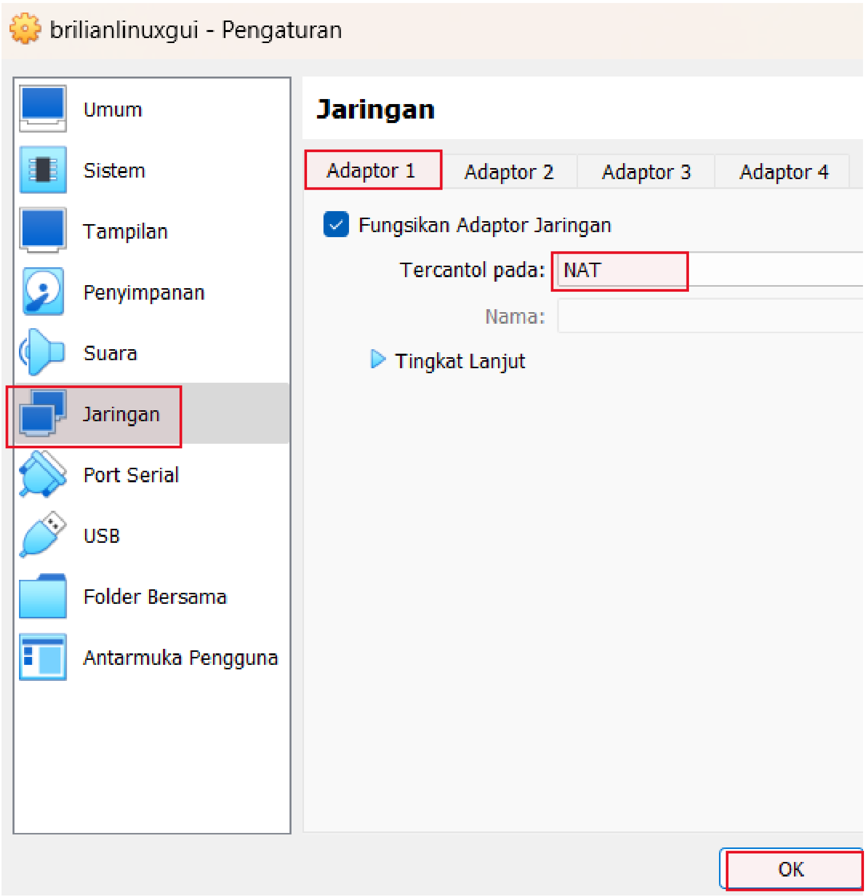
TUGAS LAPORAN PRAKTIKUM

NAMA : **BRILIAN ABEDNEGO WIRYAWAN**

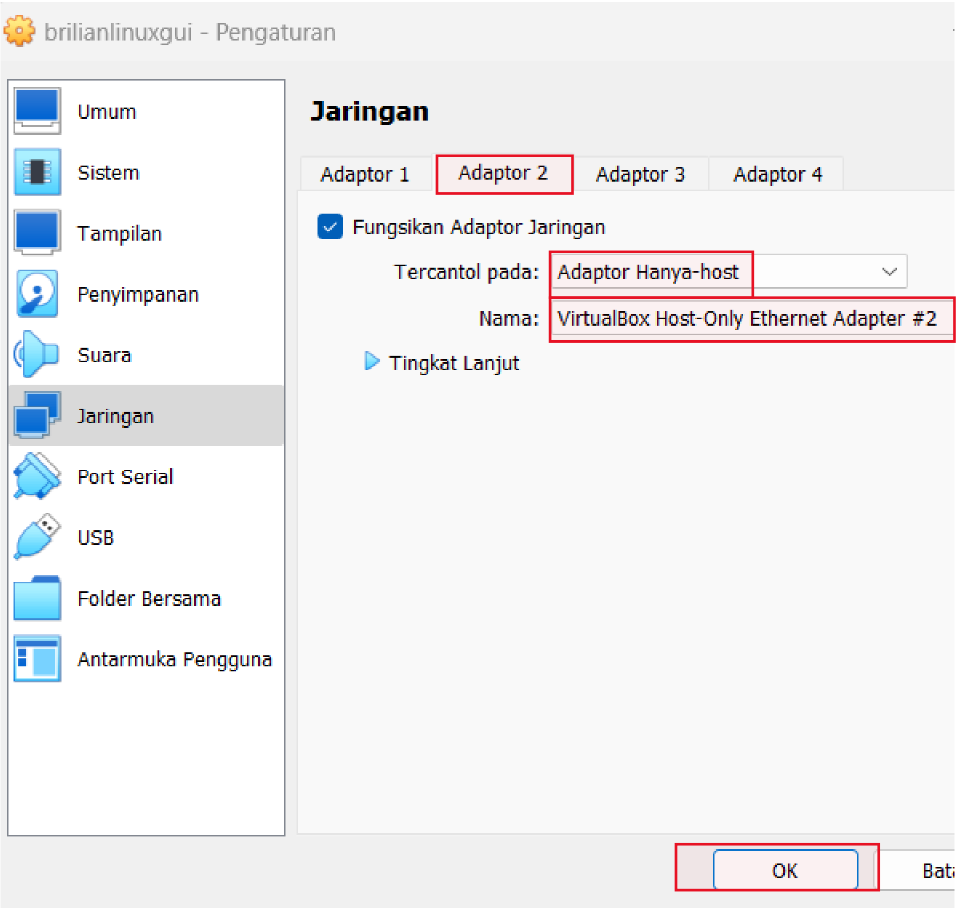
ABSEN : **06**

KELAS : **XI TJKT 1**

1. **Disini saya akan menggunakan 2 buah adapter di pengaturan sebelum start ke virtualbox Debian. Ikuti pada gambar dibawah.**



2. **Adapter yang kedua menjadi Internal Network seperti di bawah ini.**



3. **Login ke akun root dan isi password.**

```
brilian@brilian:~$ su -
Password:
root@brilian:~#
```

4. **Setelah itu atur ip kalian supaya bisa di akses oleh client. Ikuti perintah dibawah.**

```
root@brilian:~# nano /etc/network/interfaces
  GNU nano 7.2                    /etc/network/interfaces *
# This file describes the network interfaces available on your system
# and how to activate them. For more information, see interfaces(5).

source /etc/network/interfaces.d/*

# The loopback network interface
auto lo
iface lo inet loopback
 # enp0s3
auto enp0s3
iface enp0s3 inet dhcp

#bakwan goreng
allow-hotplug enp0s8
iface enp0s8 inet static
        address 10.10.6.106
        netmask 255.255.255.192
```

5. **Setelah itu restart agar bisa diperintah oleh perintah selanjutnya.**

```
root@brilian:~# systemctl restart networking
root@brilian:~#
```

6. **Lalu perintah selanjutnya lakukan update dan upgrade dengan perintah gambar di bawah. Jika ada kata hit dan gate itu tandanya berhasil.**

```
root@brilian:~# apt update && apt upgrade
Hit:1 http://deb.debian.org/debian bookworm InRelease
Hit:2 http://deb.debian.org/debian bookworm-updates InRelease
Hit:3 http://security.debian.org/debian-security bookworm-security InRe
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
20 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
```

7. **Jika sudah berhasil maka lakukan install sesuai perintah di bawah. Jika ada kata merah tidak usah takut karna memang belom di setting.**

```
root@brilian:~# apt install isc-dhcp-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required
  libdbus-glib-1-2
Use 'apt autoremove' to remove it.
The following additional packages will be installed:
  policycoreutils selinux-utils
Suggested packages:
  policykit-1 isc-dhcp-server-ldap ieee-data
Job for isc-dhcp-server.service failed because the control process exited
rror code.
See "systemctl status isc-dhcp-server.service" and "journalctl -xeu isc-d
ver.service" for details.
invoke-rc.d: initscript isc-dhcp-server, action "start" failed.
× isc-dhcp-server.service - LSB: DHCP server
     Loaded: loaded (/etc/init.d/isc-dhcp-server; generated)
     Active: failed (Result: exit-code) since Wed 2024-10-30 11:59:50 WIB
ago
       Docs: man:systemd-sysv-generator(8)
    Process: 3625 ExecStart=/etc/init.d/isc-dhcp-server start (code=exite
us=1/FAILURE)
        CPU: 22ms

Oct 30 11:59:48 brilian dhcpd[3638]: bugs on either our web page at www.i
or in the README file
Oct 30 11:59:48 brilian dhcpd[3638]: before submitting a bug.  These page
in the proper
Oct 30 11:59:48 brilian dhcpd[3638]: process and the information we find
 for debugging.
Oct 30 11:59:48 brilian dhcpd[3638]:
Oct 30 11:59:48 brilian dhcpd[3638]: exiting.
Oct 30 11:59:50 brilian isc-dhcp-server[3625]: Starting ISC DHCPv4 server
```

8. **Selanjutnya ikuti perintah gambar di bawah.**

```
root@brilian:~# nano /etc/dhcp/dhcpd.conf
```

9. **Scroll ke bawah hingga menemui bagian seperti di bawah ini. Lalu hapus pada dan sesuai kan dengan kebutuhan masing-masing.**

```
 GNU nano 7.2                    /etc/dhcp/dhcpd.conf *
# This declaration allows BOOTP clients to get dynamic addresses,
# which we don't really recommend.

#subnet 10.254.239.32 netmask 255.255.255.224 {
#  range dynamic-bootp 10.254.239.40 10.254.239.60;
#  option broadcast-address 10.254.239.31;
#  option routers rtr-239-32-1.example.org;
#}

# A slightly different configuration for an internal subnet.
subnet 10.10.6.64 netmask 255.255.255.192 {
  range 10.10.6.65 10.10.6.126;
  option domain-name-servers 10.10.6.106;
  option domain-name "internal.example.org";
  option routers 10.10.6.106;
  option broadcast-address 10.10.6.127;
  default-lease-time 600;
  max-lease-time 7200;
}


^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

10. **Ikuti perintah dibawah untuk port/client.**

```
root@brilian:~# nano /etc/default/isc-dhcp-server
```

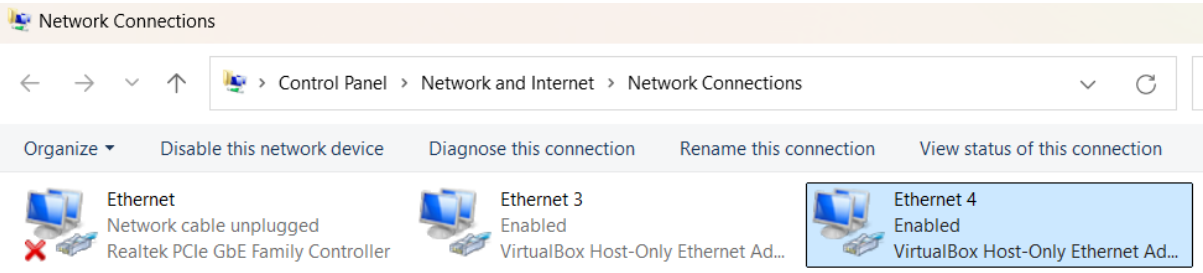
11. **Scroll kebawah dan isi ipv4 dengan “enp0s8”**

```
 GNU nano 7.2                  /etc/default/isc-dhcp-server *
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
#DHCPDv4_CONF=/etc/dhcp/dhcpd.conf
#DHCPDv6_CONF=/etc/dhcp/dhcpd6.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
#DHCPDv4_PID=/var/run/dhcpd.pid
#DHCPDv6_PID=/var/run/dhcpd6.pid

# Additional options to start dhcpd with.
#       Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
#OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
#       Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="enp0s8"
INTERFACESv6=""



^G Help       ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location
^X Exit       ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line
```

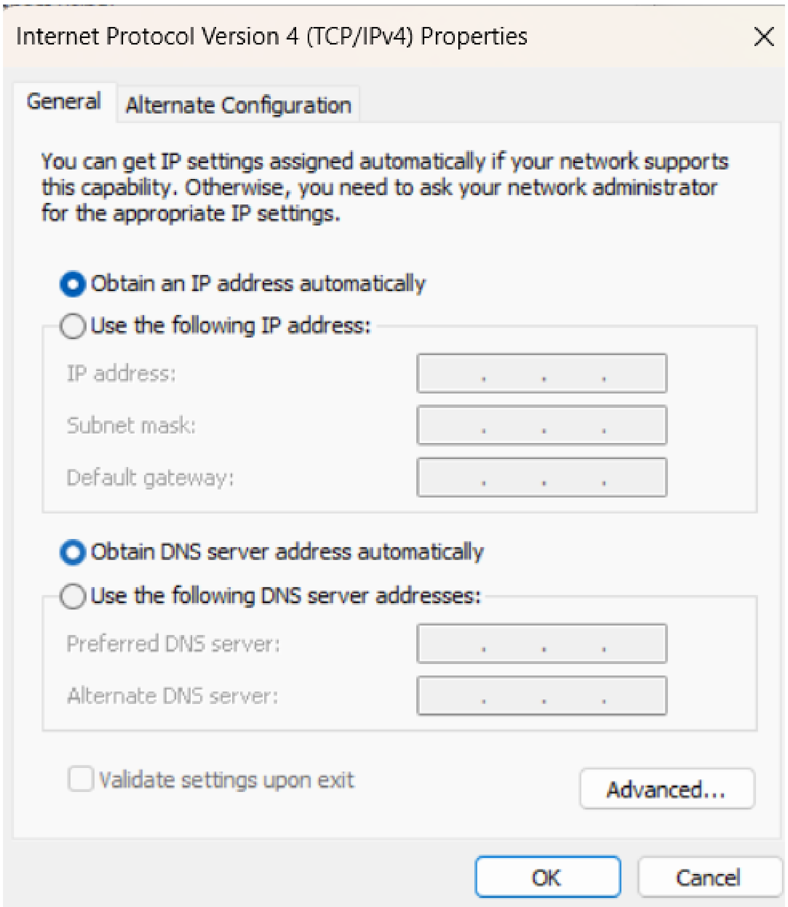
12. **Lakukan restart lagi supaya berhasil dan file yang diedit bisa di akses. Ikuti perintah di bawah.**

```
root@brilian:~# systemctl restart isc-dhcp-server
root@brilian:~#
```

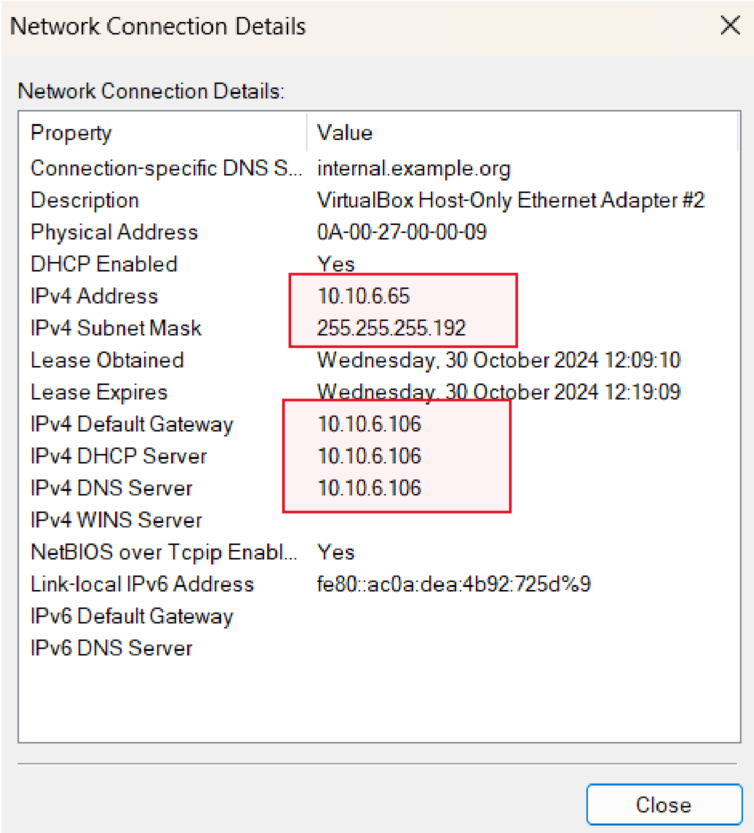
13. **Selanjutnya Control Panel –> Network And Internet -> Network Connections dan check adapter kedua.**



14. **Selanjutnya check properties dan lihat sudah dhcp atau tidak.**



15. Jika sudah check pada pada details dan check ip konfigurasi kalian.



16. **Dan bisa melakukan ping pada cmd agar memastikan bisa dipakai client.**